

DG4NLP Natural Language Processing

by Dr Amal Htait

Week 1: Coursework - Section 1

For the first section of this coursework, we will start by exploring the dataset and processing it. This foundational step involves:

- Loading and examining the data structure
- Understanding the content and format of our text data
- Applying basic preprocessing techniques
- Getting familiar with the data before moving to more advanced analysis like classification and summarization

In this practical session, we will work with a real-world dataset of scientific articles from ArXiv. You will practice basic Python skills in the context of Natural Language Processing (NLP) by preparing the data for machine learning.

The dataset source: ArXiv

What is ArXiv?

ArXiv is a free online library where researchers share their scientific papers before they are officially published. It has been around for nearly 30 years.

What's in it?

Papers from many fields: physics, computer science, math, statistics, biology, economics, and more 1.7 million articles with useful information like:

- Article titles
- Author names
- Subject categories
- Summaries (abstracts)
- Full PDFs

Why is it useful?

The collection is huge and can be overwhelming. To make it easier to work with, the creators have made the dataset available on Kaggle (<https://www.kaggle.com/datasets/Cornell-University/arxiv>). This allows researchers and students to:

- Analyse trends in research
- Build paper recommendation systems
- Predict what category a paper belongs to
- Create search tools
- Build knowledge graphs

ArXiv On Kaggle

Metadata

This dataset is a mirror of the original ArXiv data. Because the full dataset is rather large (1.1TB and growing), this dataset provides only a metadata file in the json format. This file contains an entry for each paper, containing:

- id: ArXiv ID (can be used to access the paper, see below)
- submitter: Who submitted the paper
- authors: Authors of the paper
- title: Title of the paper
- comments: Additional info, such as number of pages and figures
- journal-ref: Information about the journal the paper was published in
- doi: [<https://www.doi.org/>](Digital Object Identifier)
- abstract: The abstract of the paper
- categories: Categories / tags in the ArXiv system
- versions: A version history

Steps:

Step 1: Load and Explore the Data (from Blackboard or directly from Kaggle.com)

- How many papers are in the dataset?
- What columns/fields are available?
- What are the different categories in the categories field?
- Are there any missing values? If so, in which columns?

```
import kagglehub
```

```
# Download latest version
```

```
path = kagglehub.dataset_download("Cornell-University/arxiv")
```

```
print("Path to dataset files:", path)
```

Step 2: Handle Categories (Labels)

The categories field contains one or more subject tags for each paper. For classification, we need to assign a single main category to each paper. The list of the possible categories can be found here: https://arxiv.org/category_taxonomy

Step 3: Text Preprocessing

We will use the title and/or abstract fields as our text data for category classification. In this task, the model will learn to predict which research category (e.g., Machine Learning, Physics, Biology) a paper belongs to based on its title and/or abstract.

We will also use the abstract field to predict the title of the paper. This is a type of summarisation task, the model reads the abstract (longer text) and generates a short title that captures the main idea.

Before we can use this data for machine learning, the text must be cleaned and well organised. For example: Category labels (like "cs.CL", "physics", "math") need to be converted into numbers (0, 1, 2, 3, etc.) so the machine learning model can understand them. This is called label encoding.

Step 5: Prepare Data for Machine Learning:

- Create a csv file for the dataset, with only the following columns:
 - o Paper id
 - o Title
 - o Abstract
 - o Category (in numbers)

Coursework - Section 1 outcome:

- A dataset csv file
- Code:
 - o Well-structured and readable code used to prepare the dataset.
 - o Fully commented to explain each step of your analysis
- Documentation:
 - o Introduction:
 - The ArXiv dataset and its context
 - The purpose of this stage (initial data exploration and understanding)
 - Potential future applications (e.g. classification, search, recommendation systems)
 - o Data Exploration and Approach
A clear explanation of what you did to explore the dataset, including:
 - Understanding fields and structure
 - Handling categories and text fields
 - Any preprocessing steps applied
 - o Analysis and Visualisation:
 - Key findings from your exploration
 - At least two visualisations to support your analysis
 - Interpretation of what the visualisations show
 - o A summary of your findings and any challenges you encountered.
 - o If applicable, notes on how this analysis will support future modelling work

After completing this task, you will have a clean, processed dataset ready for the next stage: building a machine learning model to classify scientific papers by category and summarise abstracts.